Report Generated From Altium Designer

Name	Priority	Enabled	Type	Category	Scope	Attributes
AssemblyTestpoint	1	True	Assembly Testpoint Style	Testpoint	All	Under Comp - Allow Sides - Top, Bottom Pref Size = 1.524mm Pref Hole Size = 0.813mm Using Grid = Yes Grid = 0.025mm Grid Tolerance = 0mm
AssemblyTestPointUsage	1	True	Assembly Testpoint Usage	Testpoint	All	Testpoint - One Required Multiple - Not Allowed
BoardOutlineClearance	1	True	Board Outline Clearance	Manufacturing	All	Board Clearance = 0.3mm
Clearance_HV	1	True	Clearance	Electrical	InNetClass('HV') - NOT InNetClass('HV')	Clearance = 0.2mm
DiffPairsRouting_120R_Diff	1	True	Differential Pairs Routing	Routing	InDifferentialPairClass('120R_Diff')	Pref Gap = 0.44mm Min Gap = 0.126mm Max Gap = 0.44mmPref Width = 0.1mm Min Width = 0.1mm Max Width = 0.1mm
FabricationTestpoint	1	True	Fabrication Testpoint Style	Testpoint	All	Under Comp - Allow Sides - Top, Bottom Pref Size = 1.524mm Pref Hole Size = 0.813mm Using Grid = Yes Grid = 0.025mm Grid Tolerance = 0mm
FabricationTestPointUsage	1	True	Fabrication Testpoint Usage	Testpoint	All	Testpoint - One Required Multiple - Not Allowed
Fanout_BGA	1	True	Fanout Control	Routing	IsBGA	Style - Auto Direction - Alternating In and Out Via Grid = 0.025mm
Height	1	True	Height	Placement	All	Pref Height = 12.7mm Min Height = 0mm Max Height = 25.4mm
HoleSize	1	True	Hole Size	Manufacturing	All	Min = 0.2mm Max = 15mm
HoleToHoleClearance	1	True	Hole To Hole Clearance	Manufacturing	All - All	Hole To Hole Clearance = 0.254mm
LayerPairs	1	True	Layer Pairs	Manufacturing	All	Layer Pairs - Enforce
MatchedLengths_DiffPair	1	True	Matched Lengths	High Speed	InDifferentialPairClass('All Differential Pairs')	Tolerance = 0.254mm
MinimumSolderMaskSliver	1	True	Minimum Solder Mask Sliver	Manufacturing	All - All	Minimum Solder Mask Sliver = 0.04mm
NetAntennae	1	True	Net Antennae	Manufacturing	All	Net Antennae Tolerance = 0mm
PasteMaskExpansion	1	True	Paste Mask Expansion	Mask	All	Expansion = 0mm
PlaneClearance	1	True	Power Plane Clearance	Plane	All	Clearance = 0.508mm
PlaneConnect	1	True	Power Plane Connect Style	Plane	All	Style - Direct Connect
PolygonConnect	1	True	Polygon Connect Style	Plane	All - All	Style - Direct Connect
RoutingCorners	1	True	Routing Corners	Routing	All	Style - 45 Degree Min Setback = 2.54mm Max Setback = 2.54mm
RoutingLayers	1	True	Routing Layers	Routing	All	TopLayer - Enabled InternalPlane1 - DisabledMidLayer2 - Enabled InternalPlane3 - DisabledInternalPlane4 - DisabledMidLayer3 - Enabled InternalPlane2 - DisabledBottomLayer - Enabled
RoutingPriority	1	True	Routing Priority	Routing	All	Priority = 0
RoutingTopology	1	True	Routing Topology	Routing	All	Topology - Shortest
RoutingVias	1	True	Routing Via Style	Routing	All	Pref Size = 0.6mm Pref Hole Size = 0.3mm
Schematic Component Clearance Constraint_4	1	True	Component Clearance	Placement	InComponent('U46') - All	Horizontal Clearance = -50mm Vertical Clearance = -50mm
Schematic Supply Nets_18	1	True	Supply Nets	Signal Integrity	InNet('+5V_USB')	Voltage = 0.000
ShortCircuit	1	True	Short-Circuit	Electrical	All - All	Short Circuit - Not Allowed
SilkToSilkClearance	1	True	Silk To Silk Clearance	Manufacturing	All - All	Silk to Silk Clearance = 0.1mm
SilkToSolderMaskClearance	1	True	Silk To Solder Mask Clearance	Manufacturing	IsPad - All	Silk To Solder Mask Clearance = 0.1mm
SolderMaskExpansion_1	1	True	Solder Mask Expansion	Mask	InPadClass('KSZ_Pad')	Expansion = 0mm
UnpouredPolygon	1	True	Modified Polygon	Electrical	All	Allow modified - No Allow shelved - No
UnRoutedNet	1	True	Un-Routed Net	Electrical	All	(No Attributes)

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U_12V_RegB	1	True	Room Definition	Placement	InComponentClass('U_12V_RegB')	Region (BR) = (13.259mm, -58.579mm), (52.609mm, -19.204mm) Style - Keep Inside
Width_50R	1	True	Width	Routing	InNetClass('50R_Single')	Pref Impedance = 50.00ohms Min Impedance = 50.00ohms Max Impedance = 50.00ohms
Clearance	2	True	Clearance	Electrical	All - All	Clearance = 0.1mm
DiffPairsRouting_100R_Diff	2	True	Differential Pairs Routing	Routing	InDifferentialPairClass('100R_diff')	Pref Gap = 0.299mm Min Gap = 0.1mm Max Gap = 0.299mmPref Width = 0.147mm Min Width = 0.1mm Max Width = 0.147mm
Fanout_LCC	2	True	Fanout Control	Routing	IsLCC	Style - Auto Direction - Alternating In and Out Via Grid = 0.025mm
Schematic Component Clearance Constraint_10	2	True	Component Clearance	Placement	InComponent('X17') - All	Horizontal Clearance = 0mm Vertical Clearance = -25mm
Schematic Matched Net Lengths RGMII_Tx	2	True	Matched Lengths	High Speed	InNetClass('RGMII_Tx')	Tolerance = 0.127mm
Schematic Supply Nets_17	2	True	Supply Nets	Signal Integrity	InNet('+1V2_eth')	Voltage = 0.000
SolderMaskExpansion_via	2	True	Solder Mask Expansion	Mask	IsVia	Expansion = 0.102mm
U_12V_RegA	2	True	Room Definition	Placement	InComponentClass('U_12V_RegA')	Region (BR) = (51.05mm, -65.675mm), (90.3mm, -26.55mm) Style - Keep Inside
Width	2	True	Width	Routing	All	Pref Width = 0.1mm Min Width = 0.1mm Max Width = 10mm
DiffPairsRouting_90R_Diff	3	True	Differential Pairs Routing	Routing	InDifferentialPairClass('90R_Diff')	Pref Gap = 0.169mm Min Gap = 0.1mm Max Gap = 0.169mmPref Width = 0.202mm Min Width = 0.1mm Max Width = 0.202mm
Fanout_SOIC	3	True	Fanout Control	Routing	IsSOIC	Style - Auto Direction - Alternating In and Out Via Grid = 0.025mm
Schematic Component Clearance Constraint_9	3	True	Component Clearance	Placement	InComponent('X18') - All	Horizontal Clearance = 0mm Vertical Clearance = 0mm
Schematic Matched Net Lengths RGMII_Rx	3	True	Matched Lengths	High Speed	InNetClass('RGMII_Rx')	Tolerance = 0.127mm
Schematic Supply Nets_16	3	True	Supply Nets	Signal Integrity	InNet('V_SYS')	Voltage = 0.000
SolderMaskExpansion	3	True	Solder Mask Expansion	Mask	All	Expansion = 0.102mm
U_5V_RegB	3	True	Room Definition	Placement	InComponentClass('U_5V_RegB')	Region (BR) = (-36.175mm, -47.4mm), (-14.25mm, -22.625mm) Style - Keep Inside
Fanout_Small	4	True	Fanout Control	Routing	(CompPinCount < 5)	Style - Auto Direction - Out Then In Via Grid = 0.025mm
Schematic Component Clearance Constraint_8	4	True	Component Clearance	Placement	InComponent('L22_U_12V_RegA') - All	Horizontal Clearance = -50mm Vertical Clearance = -50mm
Schematic Supply Nets_15	4	True	Supply Nets	Signal Integrity	InNet('1V8_cube')	Voltage = 0.000
U_5V_Rail	4	True	Room Definition	Placement	InComponentClass('U_5V_Rail')	Region (BR) = (-80.425mm, -52.8mm), (-41.075mm, -13.475mm) Style - Keep Inside
Fanout_Default	5	True	Fanout Control	Routing	All	Style - Auto Direction - Alternating In and Out Via Grid = 0.025mm
Schematic Component Clearance Constraint_6	5	True	Component Clearance	Placement	InComponent('L22_U_12V_RegB') - All	Horizontal Clearance = -50mm Vertical Clearance = -50mm
Schematic Supply Nets_14	5	True	Supply Nets	Signal Integrity	InNet('+12V_AUX')	Voltage = 0.000
U_USB Ethernet Bridge	5	True	Room Definition	Placement	InComponentClass('U_USB Ethernet Bridge')	Region (BR) = (51.267mm, 30.2mm), (74.767mm, 58.65mm) Style - Keep Inside
Schematic Component Clearance Constraint_7	6	True	Component Clearance	Placement	InComponent('L23') - All	Horizontal Clearance = 0.254mm Vertical Clearance = -50mm
Schematic Supply Nets_13	6	True	Supply Nets	Signal Integrity	InNet('GND')	Voltage = 0.000
U_USB CAN Bridge	6	True	Room Definition	Placement	InComponentClass('U_USB CAN Bridge')	Region (BR) = (-31.25mm, 19.8mm),

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						(-15.4mm, 40.2mm) Style - Keep Inside
Schematic Component Clearance Constraint_3	7	True	Component Clearance	Placement	InComponent('X7_U_12V_RegA') - All	Horizontal Clearance = 0mm Vertical Clearance = -1.27mm
Schematic Supply Nets_12	7	True	Supply Nets	Signal Integrity	InNet('+5VB1')	Voltage = 0.000
U_5V_RegA	7	True	Room Definition	Placement	InComponentClass('U_5V_RegA')	Region (BR) = (-13.35mm, -49.775mm), (8.575mm, -25mm) Style - Keep Inside
Schematic Component Clearance Constraint_1	8	True	Component Clearance	Placement	InComponent('X7_U_12V_RegB') - All	Horizontal Clearance = 0mm Vertical Clearance = -1.27mm
Schematic Supply Nets_11	8	True	Supply Nets	Signal Integrity	InNet('+5V_SYS')	Voltage = 0.000
U_Power_Selection	8	True	Room Definition	Placement	InComponentClass('U_Power_Selection')	Region (BR) = (2.075mm, -13.45mm), (24.3mm, 14.25mm) Style - Keep Inside
Schematic Component Clearance Constraint	9	True	Component Clearance	Placement	InComponent('X10') - All	Horizontal Clearance = 0mm Vertical Clearance = -1.27mm
Schematic Supply Nets_10	9	True	Supply Nets	Signal Integrity	InNet('+2V5_eth')	Voltage = 0.000
U_EthMag_100MegB	9	True	Room Definition	Placement	InComponentClass('U_EthMag_100MegB')	Region (BR) = (-20mm, 71.4mm), (-10.25mm, 75.6mm) Style - Keep Inside
ComponentClearance	10	True	Component Clearance	Placement	All - All	Horizontal Clearance = 0.15mm Vertical Clearance = 0.15mm
Schematic Supply Nets_9	10	True	Supply Nets	Signal Integrity	InNet('+3V8_4G')	Voltage = 0.000
U_EthMag_GigB	10	True	Room Definition	Placement	InComponentClass('U_EthMag_GigB')	Region (BR) = (-25.115mm, 44.96mm), (-6.315mm, 49.21mm) Style - Keep Inside
Schematic Supply Nets_8	11	True	Supply Nets	Signal Integrity	InNet('5V_cube')	Voltage = 0.000
U_EthMag_100MegF	11	True	Room Definition	Placement	InComponentClass('U_EthMag_100MegF')	Region (BR) = (-32.5mm, 47mm), (-28.3mm, 56.75mm) Style - Keep Inside
Schematic Supply Nets_7	12	True	Supply Nets	Signal Integrity	InNet('+5V_HDMI')	Voltage = 0.000
U_EthMag_100MegE	12	True	Room Definition	Placement	InComponentClass('U_EthMag_100MegE')	Region (BR) = (-32.5mm, 64.6mm), (-28.3mm, 74.35mm) Style - Keep Inside
Schematic Supply Nets_6	13	True	Supply Nets	Signal Integrity	InNet('3V3_cube')	Voltage = 0.000
U_EthMag_100MegH	13	True	Room Definition	Placement	InComponentClass('U_EthMag_100MegH')	Region (BR) = (-32.5mm, 55.8mm), (-28.3mm, 65.55mm) Style - Keep Inside
Schematic Supply Nets_5	14	True	Supply Nets	Signal Integrity	InNet('+3V3_CR')	Voltage = 0.000
U_IO_CAN_Spray2	14	True	Room Definition	Placement	InComponentClass('U_IO_CAN_Spray2')	Region (BR) = (68.7mm, 89.4mm), (87mm, 111.2mm) Style - Keep Inside
Schematic Supply Nets_4	15	True	Supply Nets	Signal Integrity	InNet('+3V3_VDDA')	Voltage = 0.000
U_IO_CAN_FuelTank	15	True	Room Definition	Placement	InComponentClass('U_IO_CAN_FuelTank')	Region (BR) = (-25.7mm, 89.4mm), (-7.4mm, 111.2mm) Style - Keep Inside
Schematic Supply Nets_3	16	True	Supply Nets	Signal Integrity	InNet('V_FC_Primary')	Voltage = 0.000
U_EthMag_GigC	16	True	Room Definition	Placement	InComponentClass('U_EthMag_GigC')	Region (BR) = (6.411mm, 44.41mm), (25.211mm, 48.66mm) Style - Keep Inside
Schematic Supply Nets_2	17	True	Supply Nets	Signal Integrity	InNet('+12V_SYS')	Voltage = 0.000
U_EthMag_100MegC	17	True	Room Definition	Placement	InComponentClass('U_EthMag_100MegC')	Region (BR) = (28.35mm, 64.4mm), (32.55mm, 74.15mm) Style - Keep Inside
Schematic Supply Nets_1	18	True	Supply Nets	Signal Integrity	InNet('+1V8_CR')	Voltage = 0.000
U_EthMag_100MegD	18	True	Room Definition	Placement	InComponentClass('U_EthMag_100MegD')	Region (BR) = (28.35mm, 55.6mm), (32.55mm, 65.35mm) Style - Keep Inside
Schematic Supply Nets	19	True	Supply Nets	Signal Integrity	InNet('+5VA1')	Voltage = 0.000
U_EthMag_GigA	19	True	Room Definition	Placement	InComponentClass('U_EthMag_GigA')	Region (BR) = (6.889mm, 71.54mm), (25.689mm, 75.79mm) Style - Keep Inside
U_EthMag_100MegG	20	True	Room Definition	Placement	InComponentClass('U_EthMag_100MegG')	Region (BR) = (-51.45mm, 0.95mm), (-47.25mm, 10.7mm) Style - Keep Inside
U_EthMag_100MegA	21	True	Room Definition	Placement	InComponentClass('U_EthMag_100MegA')	Region (BR) = (28.35mm, 46.8mm), (32.55mm, 56.55mm) Style - Keep Inside

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U_IO_EthernetF	22	True	Room Definition	Placement	InComponentClass('U_IO_EthernetF')	Region (BR) = (8.475mm, 86.65mm), (22.375mm, 92.95mm) Style - Keep Inside
U_IO_EthernetD	23	True	Room Definition	Placement	InComponentClass('U_IO_EthernetD')	Region (BR) = (8.475mm, 95.65mm), (22.375mm, 101.95mm) Style - Keep Inside
U_IO_EthernetE	24	True	Room Definition	Placement	InComponentClass('U_IO_EthernetE')	Region (BR) = (-5.625mm, 86.65mm), (8.275mm, 92.95mm) Style - Keep Inside
U_IO_EthernetA	25	True	Room Definition	Placement	InComponentClass('U_IO_EthernetA')	Region (BR) = (-5.625mm, 104.65mm), (8.275mm, 110.95mm) Style - Keep Inside
U_IO_EthernetC	26	True	Room Definition	Placement	InComponentClass('U_IO_EthernetC')	Region (BR) = (-5.625mm, 95.65mm), (8.275mm, 101.95mm) Style - Keep Inside
U_IO_EthernetB	27	True	Room Definition	Placement	InComponentClass('U_IO_EthernetB')	Region (BR) = (8.475mm, 104.65mm), (22.375mm, 110.95mm) Style - Keep Inside
U_IO_CAN_Spray1	28	True	Room Definition	Placement	InComponentClass('U_IO_CAN_Spray1')	Region (BR) = (51mm, 89.4mm), (69.3mm, 111.2mm) Style - Keep Inside
U_FC_Power	29	True	Room Definition	Placement	InComponentClass('U_FC_Power')	Region (BR) = (-20.8mm, -11.6mm), (-14.6mm, 3.1mm) Style - Keep Inside